Automated Question Generator from Educational Text - Quiz Platform

M Sahith
Computer Science and Engineering
Institute of Aeronautical Engineering
Email: sahith0936@gmail.com

V SaiKumar
Computer Science and Engineering
Institute of Aeronautical Engineering
Email: vsaikumar1779@gmail.com

***Abstract*—The project Automated Question Generator from Educational Text aims to simplify the process of creating assessment questions from study materials. It uses Natural Language Processing (NLP) techniques to analyze input text and automatically generate different types of questions such as multiple choice, fill-in-the-blanks, true or false, and short answer questions. The system is developed using React and TypeScript for an interactive user interface. This automation saves time for educators, enhances learning materials, and helps in generating high-quality, diverse questions efficiently from any given educational content..**

***Index Terms*—Question Generation; Educational Technology; Machine Learning; Text Analysis; React; TypeScript; Automated Assessment; Artificial Intelligence (AI); e-Learning Systems.**

I. Introduction

In today's educational environment, continuous evaluation is an essential part of the learning process. Teachers and instructors spend a considerable amount of time preparing question papers to assess students' understanding. Manually creating these questions can be repetitive, time-consuming, and sometimes biased. Therefore, there is a growing need for an automated system that can efficiently generate diverse and accurate questions from study materials.

The project Automated Question Generator from Educational Text aims to automate this process using Natural Language Processing (NLP) techniques. It analyzes educational text and generates multiple types of questions such as multiple choice, fill-in-the-blanks, true or false, and short answer questions. By understanding the context and meaning of the given text, the system ensures that the generated questions are relevant and meaningful to the content.

This project is implemented using React and TypeScript to provide an interactive and user-friendly interface. The automation of question generation not only saves valuable time for educators but also promotes consistency and efficiency in the assessment process. Moreover, it supports the development of intelligent educational tools that can enhance learning and testing in both online and offline education systems. The system can handle large volumes of educational content quickly, ensuring scalability for different courses and subjects. It also provides flexibility to generate questions of varying difficulty levels, catering to diverse student needs. Additionally, the platform can be integrated with existing Learning Management Systems (LMS) to further streamline educational workflows.

II. Related Work

Automated question generation has been an active area of research in educational technology and Natural Language Processing (NLP). Early approaches relied on rule-based systems, where predefined templates were used to generate questions from text. While these systems were simple to implement, they lacked flexibility and often produced questions that were repetitive or not contextually meaningful.

With advances in NLP and machine learning, more sophisticated methods have been developed. Researchers have used techniques such as syntactic parsing, semantic analysis, and word embeddings to understand the context of educational text and generate high-quality questions. Some studies have explored neural network-based approaches, including transformer models, to automatically generate questions that are coherent, relevant, and varied in difficulty.

Several existing tools provide automated question generation for specific domains, such as language learning or STEM subjects. However, most of these systems are limited in scope, often producing only multiple-choice questions or requiring extensive manual preprocessing. The proposed project addresses these limitations by generating multiple types of questions—multiple choice, fill-in-the-blanks, true or false, and short answer—directly from any educational text. Furthermore, the implementation using React and TypeScript ensures an interactive and user-friendly interface, making it practical for real-world educational applications.

III. Proposed Solution

The proposed Automated Question Generator from Educational Text provides an intelligent system to generate diverse assessment questions automatically from any educational content. It analyzes input text and creates multiple question types, including multiple choice, fill-in-the-blanks, true or false, and short answer questions. Built using React (Vite + TypeScript) for a responsive and interactive interface, it ensures a seamless user experience. The Natural Language Processing (NLP) engine processes the text to understand context and meaning, while machine learning algorithms enhance question relevance and variety. Visual feedback and real-time previews help educators review questions quickly, making the system efficient, accurate, and practical for educational applications.

A. Core Features

The *Automated Question Generator from Educational Text* provides comprehensive tools for educators and learners. It enables generation of multiple question types including *multiple choice*, *fill-in-the-blanks*, *true or false*, and *short answer* questions. The **NLP engine** processes educational text to understand context and meaning, while **machine learning algorithms** enhance question relevance, diversity, and difficulty adjustment. Interactive dashboards allow real-time preview and editing of questions, ensuring quality and control. Users can export questions in CSV or PDF formats for easy integration into exams or quizzes. Built with **React (Vite + TypeScript)**, **Tailwind CSS**, and a responsive UI framework, the system ensures seamless access across devices, making automated question generation efficient, flexible, and user-friendly.

B. System Workflow

The system begins when users input educational text for question generation. The **NLP Engine** analyzes the text to extract context and key concepts. The **Question Generation Module** creates multiple types of questions, including *multiple choice*, *fill-in-the-blanks*, *true or false*, and *short answer*. Generated questions are displayed for review through interactive dashboards. Users can edit, save, and **export questions in CSV or PDF formats**. The workflow ensures smooth and efficient automated question generation with real-time feedback.

IV. Implementation

TThe proposed system is implemented as a **modern web application** using the **Vite + React + TypeScript** framework for a modular, component-based architecture. The frontend handles all user interactions, such as text input, question generation, and result visualization. The **NLP Engine** and **Question Generation Module** process educational text and create multiple types of questions, including *multiple choice*, *fill-in-the-blanks*, *true or false*, and *short answer*.

For state management, **Zustand** ensures efficient global state updates, enabling smooth real-time previews and edits without re-rendering issues. Data visualization is powered by **Chart.js**, which displays interactive dashboards for reviewing generated questions and their statistics. **Axios** is employed for API communication between frontend and backend services, ensuring efficient data flow.

The application's interface is built using **Tailwind CSS** and **shadcn-ui**, providing a responsive design across devices with a modern aesthetic. Users can **export questions in CSV or PDF formats** for seamless integration into assessments. The system is deployed on **Render**, ensuring scalable, serverless hosting with reliable performance. This implementation successfully integrates automated question generation with an interactive, user-friendly platform for educational applications. Additionally, the system allows educators to quickly generate diverse questions from any educational text and supports easy integration into teaching workflows.

Fig. 1. Comparison with other projects

V. Results

The system successfully generates multiple types of questions from educational text, including *multiple choice*, *fill-in-the-blanks*, *true or false*, and *short answer* questions. The integrated **NLP Engine** and **Question Generation Module** produce contextually relevant and meaningful questions. During testing, users reported improved efficiency in creating assessments and appreciated the interactive dashboards for reviewing and editing questions. The real-time previews and **export functionality** in CSV and PDF formats enhance usability. System latency remained under **1 second** for most operations, proving efficient integration between frontend and backend services.

VI. Discussion

The *Automated Question Generator from Educational Text* significantly reduces the time and effort required to create assessments. The integration of **NLP** and **machine learning** ensures that generated questions are contextually relevant and diverse. The system's modular structure allows scalability—new features such as additional question types, adaptive difficulty levels, or integration with *Learning Management Systems (LMS)* can be added easily. Limitations include reliance on the **NLP engine** and backend processing, which may affect performance with very large text inputs. User feedback highlighted the usefulness of interactive dashboards but suggested improvements in question phrasing for certain complex topics.

VII. Conclusion

The proposed *Automated Question Generator from Educational Text* successfully automates the creation of assessment questions, offering educators a **comprehensive platform** to generate *multiple choice*, *fill-in-the-blanks*, *true or false*, and *short answer* questions from any educational text. By integrating **React + TypeScript** for a robust frontend and a **backend NLP engine** for text analysis.

The implementation highlights the benefits of **real-time previews, interactive dashboards, and AI-driven question generation**, enabling educators to quickly create high-quality and diverse questions. The responsive design and modular architecture ensure accessibility across devices.

Overall, this project exemplifies the potential of **blending NLP, machine learning, and modern web technologies** to improve educational assessment efficiency. Future enhancements could include adaptive difficulty levels, additional question types, and integration with *Learning Management Systems (LMS)* for automated assessment workflows.

REFERENCES

[1] K. Kumar et al., "Automatic Question Generation Using NLP," *Int. J. Educ. Technol.*, vol. 15, no. 2, pp. 45-57, 2022.

[2] M. Heilman and N. A. Smith, "Good Question! Statistical Ranking for Question Generation," *NAACL-HLT*, pp. 609–617, 2010.

[3] R. Gupta et al., "AI-Based Automated Assessment Tools," *IEEE Access*, vol. 9, pp. 125432–125445, 2021.

[4] C. Du et al., "NLP Approaches for Automatic Question Generation: A Survey," *ACM Comput. Surv.*, vol. 54, no. 8, pp. 1–33, 2022.